

Notes: Xiong & Yu (AER, 2011)

The Chinese Warrants Bubble

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1 Big picture

- **Question.** Can we identify a field asset-price bubble when fundamentals are publicly observable and maturity is finite?
- **Empirical setting.** The paper studies China's put-warrant market from 2005–2008, when 18 put warrants traded on the Shanghai and Shenzhen exchanges.
- **Why this setting is unusually clean.** The underlying stock price, strike price, exercise ratio, and finite maturity make the warrants' fundamental values much easier to bound than the value of houses, internet stocks, or tulips.
- **Main bubble fact.** Seventeen of the 18 put warrants entered zero-fundamental periods, defined as Black–Scholes value below 0.0005 yuan. Yet they traded at positive and often very large prices.
- **Trading frenzy.** During zero-fundamental periods, the average daily turnover rate was 328 percent, meaning the average warrant changed hands more than three times per day.
- **Magnitude.** In the zero-fundamental periods, the average warrant price was 0.948 yuan, average maximum price was 2.003 yuan, and average daily volume was 1.288 billion yuan.
- **Volatility.** Average daily return volatility was 271 percent annualized, with maximum volatility averaging 1,454 percent.
- **Maturity pattern.** Prices declined gradually as expiration approached rather than crashing instantly. Turnover and volatility intensified near the last trading day.
- **Rejected explanations.** The evidence is weak for hedging demand, rational bubble models, institutional agency problems, and pure gambling/lottery preferences.
- **Supported mechanism.** The strongest explanation is the joint effect of short-sales constraints and heterogeneous beliefs. Optimistic investors can support prices because pessimists cannot short the warrants, and high turnover reflects speculative resale options.
- **Experimental-bubble link.** The setting resembles laboratory asset-market bubbles: fundamentals are observable, maturity is finite, and yet prices persistently exceed value.
- **Additional dynamics.** The paper finds positive feedback at short intraday horizons and indirect evidence that sophisticated investors may ride the bubble rather than attack it.

2 Data and measurement

2.1 Market setting

- In 2005–2008, Chinese listed companies issued put warrants with maturities between six months and two years.
- A put warrant gives the holder the right to sell the underlying stock at a predetermined strike during the exercise period.
- The 2005–2007 Chinese stock-market boom pushed the underlying stocks far above the put-warrant strike prices.
- As a result, many warrants became deep out of the money and almost surely worthless, but their market prices remained positive.
- Short sales of both the warrants and the underlying stocks were prohibited, limiting arbitrage.

Interpretation. The market is almost tailor-made for studying bubbles: the object is finite-lived, fundamentals can be bounded, and short-sales constraints prevent pessimists from correcting prices.

2.2 Data sources

- The authors obtain data on all 18 put warrants from GTA.

- The data include contract terms, contract modifications such as stock-split adjustments, daily open/high/low/close prices, daily trading volume, and intraday transaction records.
- They cross-check the data with exchange records.
- They obtain underlying stock information from CSMAR.
- For some tests, they use intraday five-minute returns to compute warrant volatility and feedback dynamics.

Interpretation. The data cover both low-frequency contractual/fundamental information and high-frequency trading behavior, allowing the paper to measure bubbles and speculative dynamics.

2.3 Black–Scholes fundamental value

The first fundamental measure is the Black–Scholes value of the put warrant:

$$BS_{i,t} = BS(S_{i,t}, K_i, T_{i,t}, \sigma_{i,t}, r_t),$$

where $S_{i,t}$ is the underlying stock price, K_i is the strike, $T_{i,t}$ is time to maturity, and $\sigma_{i,t}$ is based on the previous one-year rolling volatility of the underlying stock.

The paper defines a **zero-fundamental period** as a period in which

$$BS_{i,t} < 0.0005 \text{ yuan.}$$

Interpretation. The exact Black–Scholes number may be debatable, but a value below half the minimum trading tick means the warrant has only a tiny probability of ending in the money.

2.4 Model-free upper bound

Because Chinese stocks could not fall more than 10 percent per day, a put warrant payoff can be bounded by the lowest possible stock price before expiration. A simplified version of the bound is

$$UB_{i,t} = \max\{K_i - S_{i,t}(0.9)^{n_{i,t}}, 0\},$$

where $n_{i,t}$ is the number of trading days remaining before expiration.

Interpretation. If the observed warrant price exceeds this upper bound, the overpricing does not depend on Black–Scholes assumptions. It exceeds the maximum payoff allowed by the observed stock price and the daily price-limit rule.

2.5 Bubble sample

- The bubble sample consists of the 17 warrants that had zero-fundamental periods.
- NanHang is cash-settled and is treated separately for some calculations.
- The sample is organized by the number of trading days remaining until expiration.
- The market price during zero-fundamental periods approximates the bubble size because fundamental value is negligible.

Interpretation. The paper converts each warrant’s life into event time relative to expiration. This allows it to study maturity effects across securities with different calendar dates.

2.6 Main variables

The main empirical variables are:

$PRICE_{i,t}$: daily closing warrant price,

$TURNOVER_{i,t}$: daily warrant turnover,

$VOL_{i,t}$: annualized five-minute return volatility,
 $FLOAT_{i,t}$: total amount of warrants outstanding,
 $RET_{i,t}$: warrant return at a short intraday interval,
 $\Delta TURNOVER_{i,t}$: change in turnover over that interval.

Interpretation. These variables correspond to several bubble mechanisms: resale-option theories predict high price with high turnover, high volatility, and low float; feedback theories predict return and turnover dynamics.

3 Models and empirical specifications

3.1 Bubble definition

A warrant-price bubble exists when

$$P_{i,t} > FV_{i,t},$$

where $P_{i,t}$ is the observed price and $FV_{i,t}$ is a reasonable fundamental value.

In the zero-fundamental periods,

$$FV_{i,t} \approx 0, \quad \text{Bubble}_{i,t} \approx P_{i,t}.$$

Interpretation. This is why the paper can use the price itself as the dependent variable in bubble-size regressions.

3.2 Upper-bound violation

The model-free bubble test is

$$P_{i,t} > UB_{i,t}.$$

Interpretation. This condition is stronger than saying Black–Scholes is low. It says the price exceeds a payoff bound derived from the stock’s price-limit rule.

3.3 Jump-to-ruin premium

One possible explanation is that the warrant hedges a sudden crash in the underlying stock. The paper estimates an upper bound on this jump-risk premium.

Let P_0 be the warrant’s closing price on its last trading day, and let $T = 5$ be the number of business days in the post-trading exercise window. Define the daily jump premium:

$$J = \frac{P_0}{T}.$$

If there are t trading days remaining, the value attributable to jump risk is approximated as

$$V_t = (T + t)J.$$

The fraction of price unexplained by jump-to-ruin risk is

$$\frac{P_t - V_t}{P_t}.$$

Interpretation. Even after assigning all of the last-day warrant value to jump-risk insurance, most of the earlier warrant price remains unexplained.

3.4 Resale-option regressions

The paper tests heterogeneous-beliefs/resale-option predictions with panel regressions that include maturity fixed effects. The first regression is

$$PRICE_{i,t} = b_0 + b_1 TURNOVER_{i,t} + \text{Maturity FE} + e_{i,t}.$$

The second is

$$PRICE_{i,t} = c_0 + c_1 VOL_{i,t} + \text{Maturity FE} + e_{i,t}.$$

The float regression is

$$PRICE_{i,t} = d_0 + d_1 FLOAT_{i,t} + \text{Maturity FE} + e_{i,t}.$$

The combined specification is

$$PRICE_{i,t} = a_0 + a_1 TURNOVER_{i,t} + a_2 VOL_{i,t} + a_3 FLOAT_{i,t} + \text{Maturity FE} + e_{i,t}.$$

Interpretation. Scheinkman–Xiong-style resale-option theory predicts $a_1 > 0$, $a_2 > 0$, and $a_3 < 0$: bubble size should be higher when trading intensity and belief volatility are high, and when float is limited.

3.5 Issue-float decomposition

The paper decomposes float into warrants initially issued by the underlying firm and warrants newly issued by brokerage firms:

$$PRICE_{i,t} = q_0 + q_1 FIRMISUE_i + q_2 BROKERAGEISUE_{i,t} + \text{Maturity FE} + e_{i,t}.$$

Interpretation. Firm issuance is a fixed supply shock that increases float and should reduce the bubble. Brokerage issuance expands in response to high prices, so its coefficient can be positive despite the supply effect.

3.6 Feedback-loop regression

To test feedback trading, the paper estimates:

$$\begin{aligned} RET_{i,t} \text{ or } \Delta TURNOVER_{i,t} = & h_0 + h_1 RET_{i,t-1} + h_2 RET_{i,t-1}^+ + h_3 \Delta TURNOVER_{i,t-1} \\ & + \text{Warrant FE} + \text{Maturity FE} + e_{i,t}, \end{aligned}$$

where

$$RET_{i,t-1}^+ = \max(0, RET_{i,t-1}).$$

Interpretation. A positive h_1 in the return regression means positive feedback: past returns predict current returns. In this market, feedback appears only at very short horizons.

3.7 Momentum portfolio tests

The paper constructs winner and loser portfolios using lagged warrant returns during a formation period and then reports holding-period returns. The horizons range from five minutes to one day.

Interpretation. If smart traders could profitably ride the bubble, winner portfolios should earn positive returns. If the main profit opportunity is shorting overpriced losers but shorting is banned, loser portfolio returns should be negative but hard to exploit.

4 Identification and empirical design

4.1 What makes the bubble identification credible

- The payoff is tied to publicly observable stock prices.
- The warrants have predetermined finite maturities.
- The daily 10 percent stock-price limit gives a model-free upper bound on maximum put payoff.

- Seventeen warrants have Black–Scholes values below 0.0005 yuan for multiple days.
- Many prices exceed both Black–Scholes value and the model-free upper bound.

Interpretation. This is much cleaner than most historical bubble episodes. The key identification step is not estimating growth prospects; it is bounding the payoff of a finite-lived derivative.

4.2 Why short-sales constraints matter

- Investors could not short the warrants.
- Investors also could not short the underlying stocks to arbitrage put-warrant mispricing.
- Pessimists could therefore exit but could not force prices down by shorting.
- Optimists and speculators with resale motives could keep trading the warrant at positive prices.

Interpretation. The evidence is consistent with the classic mechanism: heterogeneous beliefs produce disagreement, and short-sales constraints prevent the pessimistic side from disciplining prices.

4.3 Why high turnover is central

- A zero-fundamental asset can still be valuable to a trader who expects to resell it to someone else.
- High turnover indicates frequent resale and belief disagreement.
- In Table 5, bubble size is strongly positively related to daily turnover.
- Turnover also rises near expiration, when the resale option becomes shorter but speculative trading becomes more intense.

Interpretation. High turnover is not a side feature. It is one of the main measurable signs that traders valued the option to resell to later optimists.

4.4 How the paper rules out competing theories

- **Hedging:** Warrant returns have small and insignificant correlation with underlying stock returns.
- **Jump-to-ruin risk:** The estimated jump-risk premium explains only a small fraction of the price.
- **Agency problems:** Institutional participation in warrants is too small to be the main driver.
- **Gambling:** Most warrant returns are negatively skewed, while lottery-preference explanations require positive skewness.
- **Rational bubbles:** Finite maturity makes standard rational bubble explanations difficult.

Interpretation. The paper does not prove one complete behavioral model, but it narrows the plausible mechanism to short-sale constraints plus heterogeneous beliefs, with feedback and naivete reinforcing the process.

4.5 What the paper cannot fully prove

- It lacks account-level investor data, so it cannot directly identify naive traders, informed traders, or manipulators.
- It cannot observe individual investor beliefs.
- It cannot fully separate belief heterogeneity from overconfidence, learning errors, or manipulation.
- It cannot directly estimate who earned trading profits during the bubble.

Interpretation. The paper is strongest on documenting the bubble and testing broad mechanisms. It is less able to identify exactly which investor types bought and sold at each point.

5 Tables and figures

The visuals below are directly cropped from the paper and grouped to keep related evidence close together.

5.1 Figure 1 + Figure 2: WuLiang as the main example



FIGURE 1. PRICES OF WULIANG PUT WARRANT

(a) WuLiang prices, Black-Scholes value, strike, and upper bound

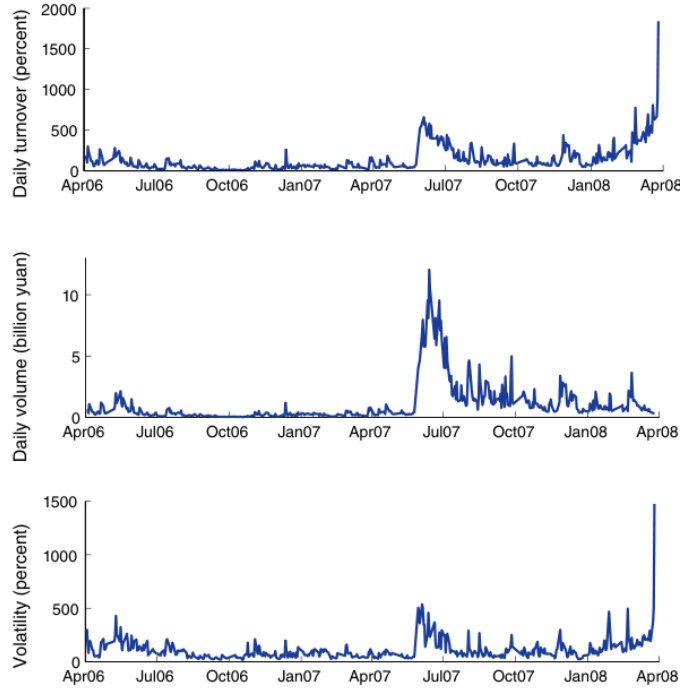


FIGURE 2. VOLUME AND VOLATILITY OF WULIANG PUT WARRANT

(b) WuLiang turnover, volume, and volatility

Read:

- Figure 1 shows that WuLiang stock rose sharply, making the put warrant deeply out of the money.
- The Black-Scholes value fell essentially to zero after July 2007, but the market price remained around several yuan for months.
- The model-free upper bound also fell to zero before expiration, while the warrant price remained positive.
- Figure 2 shows extraordinary trading intensity: turnover and volatility spike, especially near the end.
- The WuLiang warrant's last-day turnover reached 1,841 percent.

Economic message. WuLiang is a transparent bubble: value is near zero by both model-based and model-free measures, yet trading remains enormous.

5.2 Table 1 + Table 2 Panel A: contract terms and zero-fundamental trading

TABLE 1—SUMMARY INFORMATION OF THE 18 PUT WARRANTS

Name (exchange)	Warrant shares (million)	Trading period		Price at trading end			Exercise period	
		Begin	End	Stock price	Strike price	Exercise ratio	Begin	End
WanKe (SZ)	2,140	12/5/2005	8/28/2006	6.79	3.638	1	8/29/2006	9/4/2006
ShenNeng (SZ)	438	4/27/2006	10/19/2006	7.25	6.692	1	10/20/2006	10/26/2006
WuGang (SH)	474	11/23/2005	11/15/2006	3.35	2.83	1	11/16/2006	11/22/2006
JiChang (SH)	240	12/23/2005	12/15/2006	7.94	6.9	1	3/23/2006	12/22/2006
YuanShui (SH)	280	4/19/2006	2/5/2007	6.54	4.9	1	2/6/2007	2/12/2007
HuChang (SH)	568	3/7/2006	2/27/2007	25.52	13.36	1	3/6/2007	3/6/2007
BaoGang (SH)	715	3/31/2006	3/23/2007	5.7	2.37	1	3/26/2007	3/30/2007
WanHua (SH)	85	4/27/2006	4/19/2007	38.75	9.22	1.41	4/20/2007	4/26/2007
GangFan (SH)	233	12/5/2005	4/24/2007	10.72	3.16	1.535	5/8/2007	5/8/2007
HaiEr (SH)	607	5/22/2006	5/9/2007	15.79	4.29	1	5/10/2007	5/16/2007
YaGe (SH)	635	5/22/2006	5/14/2007	26.44	4.09	1	5/17/2007	5/21/2007
MaoTai (SH)	432	5/30/2006	5/22/2007	94.84	30.3	0.25	5/29/2007	5/29/2007
JiaFei (SZ)	120	6/30/2006	6/22/2007	45.21	15.1	1	6/25/2007	6/29/2007
ZhaoHang (SH)	2,241	3/2/2006	8/24/2007	39.04	5.45	1	8/27/2007	8/31/2007
ZhongJi (SZ)	424	5/25/2006	11/16/2007	24.11	7.302	1.37	11/19/2007	11/23/2007
HuaLing (SZ)	633	3/2/2006	2/22/2008	12.45	4.718	1	2/27/2008	2/29/2008
WuLiang (SZ)	313	4/3/2006	3/26/2008	25.92	5.627	1.402	3/27/2008	4/2/2008
NanHang (SH)	1,400	6/21/2007	6/13/2008	8.48	7.43	0.5	6/20/2008	6/20/2008

Notes: This table shows, for each put warrant, its name, exchange (SZ for Shenzhen, SH for Shanghai), total shares outstanding at the start of warrant trading, trading period, the closing prices of the underlying stock and the warrant on the last trading day, the exercise ratio (number of underlying stocks per warrant) on the last trading day, and the exercise period. The warrants are sorted according to the expiration date of their exercise periods.

(a) Summary information of the 18 put warrants

TABLE 2—MARKET DYNAMICS DURING THE ZERO-FUNDAMENTAL PERIOD

Panel A. Warrant price and trading volume								
Name	First day, last day	Days	Warrant price		Daily turnover (percent)		Yuan volume (million)	
			Average	Maximum	Average	Maximum	Average	Maximum
WanKe	8/3/2006, 8/28/2006	17	0.055	0.09	178	547	176	317
ShenNeng		0						
WuGang	11/6/2006, 11/15/2006	7	0.188	0.281	765	1,695	756	1,451
JiChang	12/6/2006, 12/15/2006	6	0.721	0.96	378	725	767	1,334
YuanShui	1/24/2007, 2/5/2007	9	0.379	0.562	510	1,471	711	2,589
HuChang	1/10/2007, 2/27/2007	30	0.586	0.924	264	991	751	2,162
BaoGang	2/13/2007, 3/23/2007	24	0.448	0.588	296	1,406	1,000	2,621
WanHua	1/9/2007, 4/19/2007	67	1.014	1.39	183	1,438	299	1,700
GangFan	3/13/2007, 4/24/2007	27	0.843	1.25	237	1,316	364	881
HaiEr	12/4/2006, 5/9/2007	89	0.614	0.826	96	1,072	353	1,280
YaGe	2/12/2007, 5/14/2007	55	0.544	0.749	145	972	433	1,250
MaoTai	4/16/2007, 5/22/2007	19	0.41	0.71	289	815	523	1,875
JiaFei	4/9/2007, 6/22/2007	44	1.97	6.07	363	1,741	1,285	7,990
ZhaoHang	12/18/2006, 8/24/2007	165	0.628	3.269	180	1,198	6,384	45,683
ZhongJi	5/22/2007, 11/16/2007	111	3.034	7.12	312	1,662	3,680	17,053
HuaLing	9/13/2007, 2/22/2008	40	1.768	3.87	246	1,306	1,428	3,907
WuLiang	7/23/2007, 3/26/2008	150	2.815	5.218	210	1,841	1,286	4,974
NanHang	6/11/2008, 6/13/2008	3	0.093	0.172	926	1,261	1,694	2,677
Average		48	0.948	2.003	328	1,262	1,288	5,867
Average(first 9)		21	0.529	0.756	351	1,199	603	1,632
Average(last 9)		75	1.320	3.112	307	1,319	1,896	9,632
Difference		54	0.790	2.356	-44	120	1,293	8,000
p-value		0.016	0.059	0.019	0.691	0.526	0.070	0.118

(b) Zero-fundamental price, turnover, and volume

Read:

- Table 1 lists all 18 put warrants and their contract terms.
- Maturities range from roughly six months to two years.
- Table 2 Panel A shows that 17 of 18 warrants had zero-fundamental periods.
- The average zero-fundamental period lasted 48 trading days.
- Average price in these periods was 0.948 yuan, with average maximum price of 2.003 yuan.
- Average daily turnover was 328 percent and average daily volume was 1.288 billion yuan.

Economic message. These are not thinly traded leftovers. The bubble sample is broad, active, and economically large.

5.3 Table 2 Panel B + Figure 3: volatility, bounds, and maturity dynamics

TABLE 2—MARKET DYNAMICS DURING THE ZERO-FUNDAMENTAL PERIOD (Continued)

Panel B. Warrant volatility, correlation with stock, skewness, and violation of price bound								
Name	Volatility (percent)		Correlation(stock, put)		Skewness(put ret)		Days	Average magnitude
	Average	Maximum	Correlation	p-value	Skewness	p-value		
WanKe	306	1,139	-0.424	0.090	-1.879	0.001	1	0.001
ShenNeng							0	
WuGang	763	2,297	0.045	0.925	0.931		0	
JiChang	278	414	0.383	0.454	-0.899		0	
YuanShui	397	1,362	-0.166	0.669	-0.292	0.613	0	
HuChang	214	1,309	0.46	0.011	-2.461	0	3	0.319
BaoGang	165	1,019	-0.082	0.704	-2.891	0	5	0.225
WanHua	133	1,772	-0.035	0.777	-2.757	0	9	0.527
GangFan	173	1,438	-0.435	0.023	-3.069	0	2	0.153
HaiEr	100	1,620	-0.021	0.844	-4.945	0	7	0.266
YaGe	122	1,433	0.006	0.967	-3.747	0	11	0.295
MaoTai	316	2,112	-0.142	0.561	-1.452	0.006	7	0.123
JiaFei	300	1,669	0.003	0.983	1.703	0	8	2.564
ZhaoHang	153	1,872	-0.11	0.158	4.861	0	13	0.264
ZhongJi	180	1,150	-0.052	0.587	-0.766	0.002	8	0.216
HuaLing	242	1,384	-0.013	0.938	-1.181	0.003	8	0.256
WuLiang	131	1,475	0.094	0.251	-2.954	0	16	0.261
NanHang	626	1,248	-0.887	0.306	-0.488		6	0.165
Average	271	1,454	-0.081	0.289	-1.311	0.033	5.8	0.403
Average(first 9)	304	1,344	-0.032		-1.665		2.2	0.245
Average(last 9)	241	1,551	-0.125		-0.997		9.3	0.490
Difference	-63	208	-0.093		0.668		7.1	0.245
p-value	0.501	0.357	0.550		0.555		0.000	0.395

(a) Zero-fundamental volatility, correlation, skewness, and bound violations

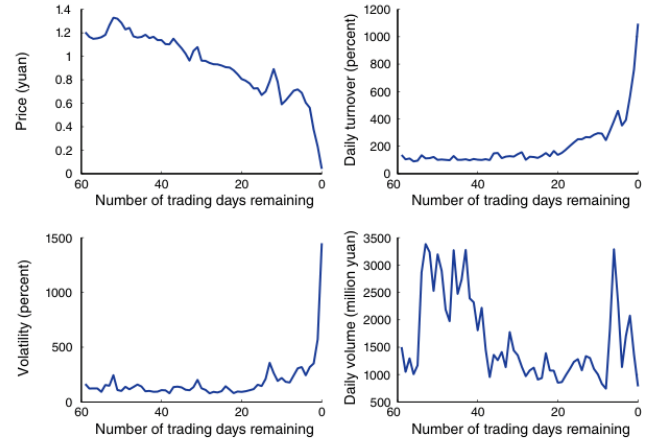


FIGURE 3. WARRANT DYNAMICS IN THE BUBBLE SAMPLE

This figure shows the average warrant price, the average daily warrant turnover, the average 5-minute volatility (annualized), and the average daily warrant trading volume (in million yuan) against the trading days remaining for the 17 put warrants in the bubble sample. The averaging is across all warrant number of trading days remaining.

(b) Average bubble-sample dynamics by days to expiration

Read:

- Table 2 Panel B reports average annualized volatility of 271 percent and average maximum volatility of 1,454 percent.
- The average correlation with the underlying stock is -0.081 and statistically insignificant.
- Average skewness is -1.311, inconsistent with a simple lottery-ticket interpretation.
- Warrants violate the model-free upper bound for an average of 5.8 days.
- Figure 3 shows that prices decline gradually as maturity approaches, while turnover and volatility rise sharply near the end.

Economic message. The bubble is not explained by hedging or lottery demand. The maturity pattern resembles a speculative resale market approaching a known terminal date.

5.4 Figure 4 + Table 3: last-day dynamics and irrational exercises

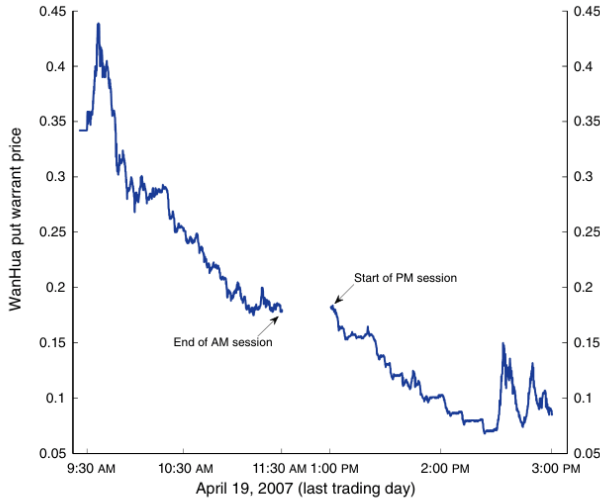


FIGURE 4. LAST-DAY PRICE DYNAMICS OF WANHUA PUT WARRANT

(a) Last-day price dynamics of WanHua

Name	Intraday price			Turnover (percent)	Volume (million)	Volatility (percent)	Exercise	
	Close	Average	Maximum				Exercised (percent)	Loss (000)
WanKe	0.001	0.011	0.019	547	98	1,139	0.08	50
ShenNeng	0.002	0.016	0.033	616	40	1,449	0.01	18
WuGang	0.06	0.172	0.3	1,529	1,240	2,297	0.01	31
JiChang	0.332	0.53	0.68	509	699	414	0.02	58
YuanShui	0.041	0.258	0.598	1,168	1,013	1,362	0.02	107
HuChang	0.015	0.209	0.442	991	1,089	1,309	0.05	38
BaoGang	0.006	0.124	0.27	1,406	1,215	1,019	0.47	111
WanHua	0.087	0.187	0.441	1,438	482	1,772	0.32	114
GangFan	0.01	0.147	0.32	1,316	422	1,438	0.27	73
HaiEr	0.001	0.049	0.11	1,072	340	1,620	0.83	581
YaGe	0.002	0.024	0.059	972	159	1,433	1.18	1,667
MaoTai	0.003	0.017	0.041	801	103	2,112	0.20	141
JiaFei	0.107	0.601	1	1,741	1,250	1,639	0	0
ZhaoHang	0.002	0.069	0.176	968	3,036	1,872	0.02	163
ZhongJi	0.01	0.076	0.107	1,662	469	1,150	0	0
HuaLing	0.01	0.093	0.15	1,306	648	1,340	0	0
WuLiang	0.004	0.054	0.08	1,841	285	1,475	0	0
NanHang	0.003	0.044	0.099	1,261	793	1,248	0	0
Average	0.039	0.149	0.274	1,175	743	1,449	0.19	175
Average(first 9)	0.062	0.184	0.345	1,058	700	1,355	0.14	67
Average(last 9)	0.016	0.114	0.202	1,292	787	1,543	0.25	284
Difference	-0.046	-0.070	-0.142	234	87	188	0.11	217
p-value	0.234	0.397	0.274	0.225	0.802	0.361	0.504	0.257

Notes: This table reports, for each warrant on its last trading day, the close/average/maximum of its intraday transaction price, the daily turnover, trading volume (in million yuan), and 5-minute warrant return volatility (annualized). Also reported are the fraction of warrant exercised and the total exercise loss (in thousand yuan, assuming the stock price at exercise equals the closing price on the last trading day). Cross-sectional averages for all warrants, for the first 9 warrants, and for the last 9 warrants in the full sample are reported along with the differences between the averages of the first 9 and the last 9 warrants and the p-values for testing the null hypothesis of no difference

(b) Last trading day and irrational warrant exercises

Read:

- Figure 4 shows that WanHua's last trading day did not crash immediately. It declined with repeated intraday runups.
- Table 3 shows the average last-day close was 0.039 yuan, but the average intraday price was 0.149 yuan and the average maximum was 0.274 yuan.
- Average last-day turnover was 1,175 percent, and average last-day volatility was 1,449 percent.
- Thirteen warrants had irrational exercises by investors, producing positive exercise losses.

Economic message. Even at the terminal date, when fundamental value was essentially zero, speculation continued and some investors made dominated exercise decisions.

5.5 Table 4 + Table 5: testing jump risk and resale-option predictions

TABLE 4—FRACTION OF WARRANT PRICE UNEXPLAINED BY JUMP-TO-RUIN RISK

Remaining trading days	Fraction of warrant price unexplained by jump risk					Warrant last close	Jump premium
	1	2	3	4	5		
GangFan	0.96	0.97	0.98	0.98	0.97	0.95	0.002
WanKe	0.94	0.96	0.96	0.95	0.95	0.96	0.001
HuaLing	0.92	0.92	0.91	0.91	0.90	0.85	0.001
WuLiang	0.94	0.96	0.97	0.97	0.97	0.97	0.004
ZhongJi	0.89	0.91	0.91	0.90	0.93	0.95	0.01
JiaFei	0.76	0.89	0.96	0.95	0.96	0.90	0.0214
MaoTai	0.93	0.96	0.97	0.97	0.98	0.98	0.003
HaiEr	0.99	0.99	0.99	0.99	0.99	0.99	0.001
YaGe	0.96	0.98	0.99	0.99	0.99	0.99	0.002
WanHua	0.71	0.83	0.79	0.78	0.79	0.61	0.087
YuanShui	0.89	0.80	0.81	0.80	0.80	0.79	0.041
BaoGang	0.98	0.98	0.98	0.97	0.97	0.96	0.006
HuChang	0.96	0.96	0.95	0.94	0.93	0.90	0.015
ZhaoHang	0.98	0.99	0.99	0.98	0.99	0.96	0.002
JiChang	0.35	0.28	0.38	0.32	0.28	-0.09	0.332
WuGang	0.36	-0.47	0.36	0.53	0.56	0.35	0.06
Average	0.84	0.81	0.87	0.87	0.87	0.81	0.043
t -stat for fraction = 0	16.44	8.43	17.08	18.42	18.00	11.03	

Notes: This table shows the fraction of warrant price unexplained by jump risk premium. For each warrant, the jump risk premium per day, denoted J , is set to its price at the close of its last trading day (denoted by P_0) divided by the number of business days after the last trading day but before the expiration of warrant exercise (denoted by T), i.e., $J = P_0/T$. $T = 5$ for all the warrants in the sample. Then, the warrant price attributable to jump risk during a warrant trading day is computed as $V_t = (T + t)J$ where t is the number of trading days remaining. The fraction of warrant price unexplained by such jump risk premium is shown for each warrant (excluding NanHang and ShenNeng) when there are up to 10 trading days remaining. The fraction is computed as $(P_t - V_t)/P_t$, where P_t is the warrant

TABLE 5—DETERMINANTS OF SIZE OF THE WARRANTS BUBBLE

	(1)	(2)	(3)	(4)	(5)	(6)
<i>TURNOVER</i>	0.203			0.158		0.105
(t -stat)	(8.91)			(5.46)		(3.80)
<i>VOL</i>		21.76		14.12		13.99
(t -stat)		(5.38)		(2.97)		(3.05)
<i>FLOAT</i>			-0.302	-0.324		
(t -stat)			(12.62)	(12.31)		
<i>FIRMISSUE</i>					-1.302	-1.260
(t -stat)					(25.03)	(23.53)
<i>BROKERAGEISSUE</i>					0.496	0.419
(t -stat)					(15.84)	(12.60)

Notes: This table includes the regression results of daily warrant closing prices on *TURNOVER* (daily warrant turnover) in column 1, on *VOL* (5min warrant return volatility) in column 2, on *FLOAT* (the total number of warrants outstanding, in billions) in column 3, on *TURNOVER*, *VOL*, and *FLOAT* together in column 4, on *FIRMISSUE* (the total amount of warrants issued by the underlying firm from Table 1, in billions) and *BROKERAGEISSUE* (net total warrant issued by brokerage firms, in billions) in column 5, and on *TURNOVER*, *VOL*, *FIRMISSUE*, and *BROKERAGEISSUE* together in column 6. The regressions use the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike) and include warrant maturity fixed effect. The t -statistics, in the parentheses, are adjusted for heteroskedasticity and correlation within a trading day.

(a) Fraction of warrant price unexplained by jump-to-ruin risk

(b) Determinants of bubble size

Read:

- Table 4 shows that jump-to-ruin risk leaves most warrant value unexplained.
- For the average warrant, 84 percent of price one day before expiration and 81 percent ten days before expiration remain unexplained by jump risk.
- Table 5 shows that warrant price is positively related to turnover: coefficient 0.203 with $t = 8.91$.
- Warrant price is positively related to volatility: coefficient 21.76 with $t = 5.38$.
- Warrant price is negatively related to float: coefficient -0.302 with $t = 12.62$.
- In the combined regression, turnover and volatility remain positive while float remains negative.

Economic message. Jump-risk insurance is too small. The strongest evidence supports a resale-option mechanism: bubble size rises with trading intensity and volatility, and falls with float.

5.6 Figure 5 + Tables 6–7: price drops, feedback, and momentum profits

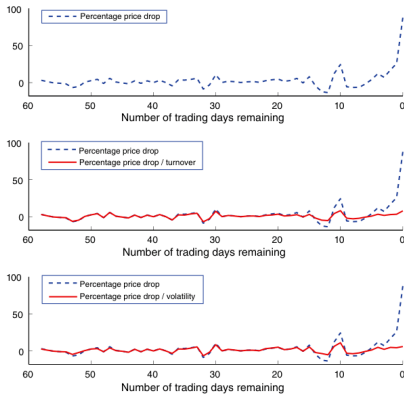


FIGURE 5. AVERAGE WARRANT PRICE DROP

(a) Average warrant price drop

Sampling frequency	<i>RET</i>				Δ <i>TURNOVER</i>			
	5 minutes	10 minutes	30 minutes	1 hour	5 minutes	10 minutes	30 minutes	1 hour
Lag <i>RET</i>	0.137 (2.42)	0.193 (2.94)	-0.036 (0.36)	-0.090 (0.68)	1.148 (3.48)	1.157 (4.59)	0.947 (2.84)	1.298 (3.98)
Lag Δ <i>RET</i>	-0.103 (1.28)	-0.115 (1.39)	0.154 (1.26)	0.224 (1.50)	-1.390 (2.13)	-1.404 (3.74)	-1.039 (3.57)	-1.822 (4.83)
Lag Δ <i>TURNOVER</i>	0.000 (0.65)	0.001 (0.96)	-0.001 (0.72)	0.002 (0.45)	-0.145 (9.62)	-0.146 (9.72)	-0.154 (11.60)	-0.124 (8.30)

Notes: This table shows the regression results of *RET* and Δ *TURNOVER* on lagged *RET*, Δ *RET*, and Δ *TURNOVER*. *RET* is warrant return, Δ *RET* = $\max(0, RET)$. Δ *TURNOVER* is the proportional change in warrant turnover. The regressions use the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike). The regressions include warrant fixed effect and maturity fixed effect. The sampling frequency goes from 5 minutes to 1 hour. The t -statistics in parentheses are adjusted for heteroskedasticity and correlation within trading day.

(b) Feedback dynamics

Return horizon	<i>WINNER</i>					<i>LOSER</i>				
	5 minutes	10 minutes	30 minutes	1 hour	2 hours	5 minutes	10 minutes	30 minutes	1 hour	2 hours
Winner	-0.0005 (1.89)	-0.0001 (0.14)	-0.0013 (0.76)	-0.0026 (0.96)	-0.0033 (0.69)	-0.0058 (0.68)	-0.0004 (1.35)	-0.0015 (2.71)	-0.0029 (2.33)	-0.0058 (2.42)
Loser	-0.0004 (1.35)	-0.0015 (2.71)	-0.0029 (2.33)	-0.0058 (2.42)	-0.0110 (2.69)	-0.0179 (2.36)				

Notes: This table shows the ex post returns of warrant portfolios sorted on lagged returns for warrants with the same number of trading days remaining during the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike). Ex post returns are reported for winner and loser warrants (defined by lagged returns being above/below median). The return horizon ranges from 5 minutes to 2 hours. The t -statistics in parentheses are adjusted for heteroskedasticity and correlation within trading day.

(c) Momentum profits

Read:

- Figure 5 shows that the average price drop accelerates during the final trading days.
- When the price drop is scaled by turnover or volatility, the maturity pattern is much flatter.
- Table 6 shows positive feedback in returns at five-minute and ten-minute horizons.
- Lagged returns also predict turnover changes at multiple horizons.
- Table 7 shows that winner portfolios do not earn positive, significant returns.
- Loser portfolios earn negative returns, which would be profitable to short, but shorting was not allowed.

Economic message. The market has very short-run feedback and unexploited shorting opportunities. This reinforces the importance of short-sales constraints.

6 Short summary

- Xiong and Yu study 18 Chinese put warrants from 2005–2008.
- The underlying stock prices, strikes, exercise ratios, and finite maturities make fundamentals unusually observable.
- Seventeen warrants had periods when Black–Scholes value was below 0.0005 yuan.
- Despite near-zero fundamentals, the warrants traded at positive and often high prices.
- The average zero-fundamental period lasted 48 trading days.
- During those periods, average price was 0.948 yuan and average turnover was 328 percent per day.
- Many prices also violated a model-free upper bound based on China’s 10 percent daily stock-price drop limit.
- Prices declined gradually as maturity approached rather than collapsing immediately.
- Turnover and volatility rose sharply near expiration.
- Hedging demand is implausible because warrant-stock return correlations are small.
- Jump-to-ruin risk explains only a small fraction of prices.
- Gambling behavior is implausible because warrant returns are mostly negatively skewed.
- Institutional agency problems are unlikely because institutions had little presence in this market.
- The strongest explanation is heterogeneous beliefs combined with short-sales constraints.
- Bubble size is positively related to turnover and volatility and negatively related to float.
- Short-run positive feedback exists in intraday returns and turnover.
- Momentum strategies do not generate positive winner returns, but loser portfolios earn negative returns that cannot be shorted.
- The paper connects a real-world bubble to experimental bubble findings: even finite-lived assets with observable fundamentals can trade above value.

7 Conclusion

7.1 Core conclusion

The paper documents a remarkably clean field bubble. Chinese put warrants with publicly observable, near-zero fundamentals continued to trade at positive prices and enormous volume. The evidence points away from hedging, rational bubbles, jump risk, agency problems, and lottery demand, and toward short-sales constraints interacting with heterogeneous beliefs and speculative resale motives.

7.2 Economic intuition

The key intuition is that a worthless asset can trade at a positive price if buyers believe they can resell it to someone more optimistic. If pessimists cannot short, they cannot force prices back to fundamental value. High turnover then becomes both a symptom and a driver of the bubble: the asset is valuable not because of terminal payoff, but because of resale opportunities before the terminal date.

7.3 Takeaway

Xiong and Yu's paper is important because it shows a bubble in a setting where fundamentals are unusually transparent. It therefore strengthens the case that bubbles are not only valuation mistakes caused by hard-to-measure fundamentals. Even when value is almost observable and maturity is finite, short-sales constraints, belief disagreement, and speculative trading can sustain large price distortions.